

Maximum reward reinforcement learning: A non-cumulative reward criterion

K.H. Quah, Chai Quek *

*Centre for Computational Intelligence, School of Computer Engineering, Nanyang Technological University,
Blk N4 #2A-32, Nanyang Avenue, Singapore 639798*

Abstract

Existing reinforcement learning paradigms proposed in the literature are guided by two performance criteria; namely: the expected cumulative-reward, and the average reward criteria. Both of these criteria assume an inherently present cumulative or additivity of the rewards. However, such inherent cumulative of the rewards is not a definite necessity in some contexts. Two possible scenarios are presented in this paper, and are summarized as follows. The first concerns with learning of an optimal policy that is further away in the existence of a sub-optimal policy that is nearer. The cumulative rewards paradigms suffer from slower convergence due to the influence of accumulating the lower rewards, and take time to fade away the effect of the sub-optimal policy. The second scenario concerns with approximating the supremum values of the payoffs of an optimal stopping problem. The payoffs are non-cumulative in nature, and thus the cumulative rewards paradigm is not applicable to resolve this. Hence, a non-cumulative reward reinforcement-learning paradigm is needed in these application contexts. A maximum reward criterion is proposed in this paper, and the resulting reinforcement-learning model with this learning criterion is termed the *maximum reward reinforcement learning*. The maximum reward reinforcement learning considers the learning of non-cumulative rewards problem, where the agent exhibits a maximum reward-oriented behavior towards the largest rewards in the state-space. Intermediate lower rewards that lead to sub-optimal policies are ignored in this learning paradigm. The maximum reward reinforcement learning is subsequently modeled with the FITSK-RL model. Finally, the model is applied to an optimal stopping problem with a nature of non-cumulative rewards, and its performance is encouraging when benchmarked against other model.

© 2005 Elsevier Ltd. All rights reserved.

Keywords: Maximum reward; Reinforcement learning; Non-cumulative reward; FITSK-RL; Optimal stopping problem; Financial derivative pricing; Two-cycle task; Maximum reward-oriented behaviour

1. Introduction

Reinforcement learning is a trial and error learning method, where the agent tries to explore and maximize the numerical state values from the feedback it receives over the long run. The process that the agent follows is modeled as the *Markov decision process* (MDP). In the MDP context, the two most well-studied optimality criteria are the expected cumulative reward and the average reward criteria (Puterman, 1994). The expected cumulative reward criterion has been adopted as the de facto objective measure in reinforcement learning research (Bertsekas, & Tsitsiklis, 1996; Kaelbling, Littman & Moore, 1996; Rummery & Niranjan, 1994; Sutton, 1984, 1988; Sutton & Barto, 1998; Watkins, 1989). This criterion employs a

technique to exponentially discount the future rewards, such that the near-future rewards are more valuable than far-future rewards. However, some research has demonstrated that the expected cumulative reward criterion is unsuitable for some problems, particularly in the undiscounted reward domain (Schwartz, 1993). Some reinforcement learning methods based on the average reward criterion have been proposed to resolve the undiscounted reward reinforcement learning problem (Mahadevan, 1996; Schwartz, 1993; Tadepalli & Ok 1998). In the essence of this, both classes of methods employ some form of cumulative rewards criteria; the former with a reward discounting approach such that their infinite cumulative rewards are finite, and the latter with a reward averaging approach such that both short and long-term rewards are indistinguishable.

However, a central issue is a problem of whether such a cumulative rewards formulation is a definite necessary requirement in all-learning contexts. Is there any problem that the cumulative rewards formulation is unnecessary, and unjustifiable? This paper discusses two possible scenarios

* Corresponding author. Tel.: +65 6790 4926; fax: +65 6792 6559.

E-mail address: ashcquek@ntu.edu.sg (C. Quek).

where the cumulative reward formulation is to be avoided for superior learning result. This is demonstrated in Sections 3 and 4. In Section 3, a two-cycle task from (Mahadevan, 1996) is discussed. One of the cycle leads to a sub-optimal policy and the other cycle leads to an optimal policy, but the rewards of the optimal policy is further away. Therefore, the focus of the learning is on how to bypass the sub-optimal policy in the search for an optimal policy. The classical Q-learning (Watkins, 1989) was demonstrated to suffer from this problem because it was being dragged down by the sub-optimal policy, causing it to be lagging behind from switching to a better policy. In Section 4, an optimal stopping problem arising from financial derivative pricing (Tsitsiklis & Van Roy, 1999) is discussed. This is a case where the cumulative rewards formulation is unjustifiable. On the other hand, this problem is concerned with finding the supremum of the payoff values of each state, which is a function of the expected discounted maximum rewards in a non-cumulative manner. Furthermore, a discounted reward is significant in this domain as the discounting operation represents a compensation of future rewards with respect to the drift rate of financial pricing. This represents a contrary argument against the claim by (Schwartz, 1993) that discounting of rewards is not necessarily important. Having these, a reinforcement learning formulation with non-cumulative *maximum reward* criterion is proposed to accommodate the learning needs of such tasks.

The maximum reward reinforcement-learning agent exhibits a maximum reward-oriented behavior. **It attempts to search for the policy that leads to the largest reward;** regardless of intermediate lower rewards along the path. This positions it to have superior immunity to the existence of sub-optimal policies in the problem domain, such as the two cycles problem in Section 3. Furthermore, the maximum reward learning is demonstrated to be effective in approximating the future payoff in a financial pricing context. The learning system exhibits an asymptotic behavior of approximating the supremum value with the maximum reward formulation, as will be discussed in Section 4.

This paper is organized as follows. Section 2 describes the general formulation of the maximum reward reinforcement learning and its update formulae as part of the modular design of the earlier proposed generic reinforcement-learning framework from (Quah & Quek, submitted for publication). Section 3 studies the learning behavior of the maximum reward reinforcement learning using a two-cycle task, and compares against two existing reinforcement-learning models. Section 4 applies the proposed maximum reward reinforcement learning with its generic learning framework to an optimal stopping problem arising from the financial derivative pricing context. Section 5 concludes this paper.

2. Maximum reward reinforcement learning

This section formulates the maximum reward reinforcement learning. Section 2.1 provides some basic definitions from the existing research. Section 2.2 mathematically formulates the maximum reward reinforcement learning, discusses the three

existing reinforcement-learning models, and the equivalent form of these three models for the maximum reward reinforcement learning. Section 2.3 solves the modelling problem of the maximum reward reinforcement learning. This is made possible by mapping the maximum reward error update formula to a generic reinforcement-learning framework to inherit the modularity and functionality of the framework.

2.1. Background definitions

Consider a sequence of an infinite horizon *Markov decision process* (MDP) (Puterman, 1994) with discrete time step $t = 0, 1, 2, 3, \dots, \infty$. At each discrete time step, the agent receives some observable environment's states, $S_t \in S$, where S is the set of possible states. The agent implements a mapping from the state to probabilities of selecting each possible action, $a_t \in A(s_t)$, where $A(s_t)$ is the set of actions available in state s_t . The mapping is called the agent's policy, π_t , and is depicted by Eq. (2.1).

$$\text{Policy mapping; } \pi_t(s, a) = \Pr(a_t = a | s_t = s) \quad (2.1)$$

indicates the probability of choosing action a when the perceived state is s at time t . Due to the action a_t , the agent receives a numerical reward r_{t+1} , and observes a new state s_{t+1} at time $t+1$.

The goal of the learning algorithm is to maximize the successive total reward, or the *return* value as described in Eq. (2.2).

$$\text{Return; } R_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} \dots = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \quad (2.2)$$

The discount rate, γ , determines the present value of future rewards, a reward received k time steps in the future is worth only γ^{k-1} times what it would be worth if it were received immediately. If $\gamma = 0$, then the agent's actions only influence the immediate reward, r_{t+1} , as in most supervised learning agent. As γ approaches 1, the agent will take future rewards more seriously, or become more farsighted.

2.2. Formulation of maximum reward reinforcement learning

This section defines a possible variation of return value, and extends the definition to various Bellman's optimality formulae. This is followed by the formulation of the maximum reward form of temporal difference learning models in reinforcement learning.

The return value is defined as the cumulative future rewards in Eq. (2.2) where the agent is learning to maximize the return value. However, if the objective of the agent is to identify the largest reward within the value space, then the return value formulae in Eq. (2.2) can be suitably modified into Eq. (2.3).

$$R_t = \max(r_{t+1}, \gamma r_{t+2}, \gamma^2 r_{t+3}, \dots) = \max_{\forall k \in [0, \infty]} (\gamma^k r_{t+k+1}) \quad (2.3)$$

The maximum reward agent will attempt to discover the shortest path that lead to the highest reward, without

concerning too much on the intermediate lower rewards. This constructs the fundamental behavior exhibited in the maximum reward reinforcement learning.

Given any state, s , and action, a , the probability of each possible next state, s' , or the *transition probability* from state s to state s' is given by Eq. (2.4).

$$P_{b'}^a = \Pr(s_{t+1} = s' | s_t = s, a_t = a) \quad (2.4)$$

Similarly, the expected value of the next reward is given by Eq. (2.5).

$$R_{ss'}^a = E(r_{t+1} | s_t = s, a_t = a, s_{t+1} = s') \quad (2.5)$$

where $E(\cdot)$ is the expectation function.

Given a policy, π , which maps state s to an action a , the *value* of the state s under policy π , denoted by $V^\pi(s)$, is defined as the expected *return* at state s by following the same policy in all subsequent states. This is described in Eq. (2.6).

$$\begin{aligned} V^\pi(s) &= E_\pi(R_t | s_t = s) = E_\pi \left(\max_{\forall k \in [0, \infty]} (\gamma^k r_{t+k+1}) | s_t = s \right) \\ &= E_\pi \left(\max \left(r_{t+1}, \gamma \max_{\forall k \in [0, \infty]} (\gamma^k r_{t+k+2}) \right) | s_t = s \right) \\ &= \sum_{\forall a \in A(s)} \pi(s, a) \sum_{\forall s' \in S'} P_{ss'}^a [\max(R_{ss'}^a, \gamma V^\pi(s'))] \end{aligned} \quad (2.6)$$

where $E_\pi(\cdot)$ is the expectation function with a given policy π .

Similarly, the *action-value function for policy π* is defined as the value of taking action a in state s under a policy π , denoted as $Q^\pi(s, a)$, and described by Eq. (2.7).

$$\begin{aligned} Q^\pi(s, a) &= E_\pi(R_t | s_t = s, a_t = a) \\ &= E_\pi \left(\max_{\forall k \in [0, \infty]} (\gamma^k r_{t+k+1}) | s_t = s, a_t = a \right) \\ &= E_\pi \left(\max \left(r_{t+1}, \gamma \max_{\forall k \in [0, \infty]} (\gamma^k r_{t+k+2}) \right) | s_t = s, a_t = a \right) \\ &= \sum_{\forall s' \in S'} P_{ss'}^a [\max(R_{ss'}^a, \gamma V^\pi(s'))] \\ &= \sum_{\forall s' \in S'} P_{ss'}^a \left[\max \left(R_{ss'}^a, \gamma \sum_{\forall a' \in A'(s')} \pi(s', a') Q^\pi(s', a') \right) \right] \end{aligned} \quad (2.7)$$

Consider an *optimal policy*, π^* , where,

$$V^{\pi^*}(s) \geq V^\pi(s), \quad (2.8)$$

for all s and π .

An optimal policy produces the largest state value for all s among all the policies. Define the *optimal state-value function*, V^* , as Eq. (2.9).

$$\begin{aligned} V^*(s) &= \max_{\pi} V^\pi(s) = \max_a Q^{\pi^*}(s, a) \\ &= \max_a \sum_{\forall s' \in S'} P_{ss'}^a [\max(R_{ss'}^a, \gamma V^*(s'))] \end{aligned} \quad (2.9)$$

Similarly, the *optimal action-value function*, Q^* , is described by Eq. (2.10).

$$\begin{aligned} Q^*(s, a) &= \max_{\pi} Q^\pi(s, a) \\ &= E(\max(r_{t+1}, \gamma V^*(s_{t+1})) | s_t = s, a_t = a) \\ &= E \left(\max \left(r_{t+1}, \gamma \max_{a'} Q^*(s_{t+1}, a') \right) | s_t = s, a_t = a \right) \\ &= \sum_{\forall s' \in S'} P_{ss'}^a \left[\max(R_{ss'}^a, \gamma \max_{a'} Q^*(s', a')) \right] \end{aligned} \quad (2.10)$$

Eqs. (2.9) and (2.10) represent the maximum reward Bellman optimality equations for V^* and Q^* .

The temporal difference error equations for the three popular reinforcement learning models are listed in Eqs. (2.12), (2.13) and (2.14). These models are the temporal difference learning (Sutton, Sutton & Barton, 1988, 1998), the SARSA learning (Rummery & Niranjan, 1994), and the Q-learning (Watkins, 1989), respectively.

The general TD learning error follows from the error function in Eq. (2.11).

$$\delta_t = R_t - E(R_t) \quad (2.11)$$

Therefore, the TD error of temporal difference learning for a state at time t is given in Eq. (2.12).

$$\delta_t = r_{t+1} + \gamma V_t(s_{t+1}) - V_t(s_t) \quad (2.12)$$

For on-policy SARSA learning, the TD error is described in Eq. (2.13).

$$\delta_t = r_{t+1} + \gamma Q_t(s_{t+1}, a_{t+1}) - Q_t(s_t, a_t) \quad (2.13)$$

Similarly, for off-policy Q-learning, the TD error is described in Eq. (2.14).

$$\delta_t = r_{t+1} + \gamma \max_a Q_t(s_{t+1}, a) - Q_t(s_t, a_t) \quad (2.14)$$

Each popular reinforcement-learning model can be modified into a respective form of maximum reward reinforcement learning model. It can be formulated by modifying the return value to take on the maximum reward, as shown in Eq. (2.3). Following these, the TD error formulae for maximum reward reinforcement learning models are shown in Eq. (2.15). The subscript M denotes that the objectives of these learning models are the maximum reward values instead of the cumulative reward values.

$$\text{TD}_M \text{ learning : } \delta_t = \max(r_{t+1}, \gamma V_t(s_{t+1})) - V_t(s_t)$$

$$\text{SARSA}_M \text{ learning : } \delta_t$$

$$= \max(r_{t+1}, \gamma Q_t(s_{t+1}, a_{t+1})) - Q_t(s_t, a_t) \quad (2.15)$$

$$\text{Q}_M - \text{learning : } \delta_t$$

$$= \max(r_{t+1}, \gamma \max_a Q_t(s_{t+1}, a)) - Q_t(s_t, a_t)$$

Further extensions of these maximum reward TD error formulae to the respective reinforcement learning models

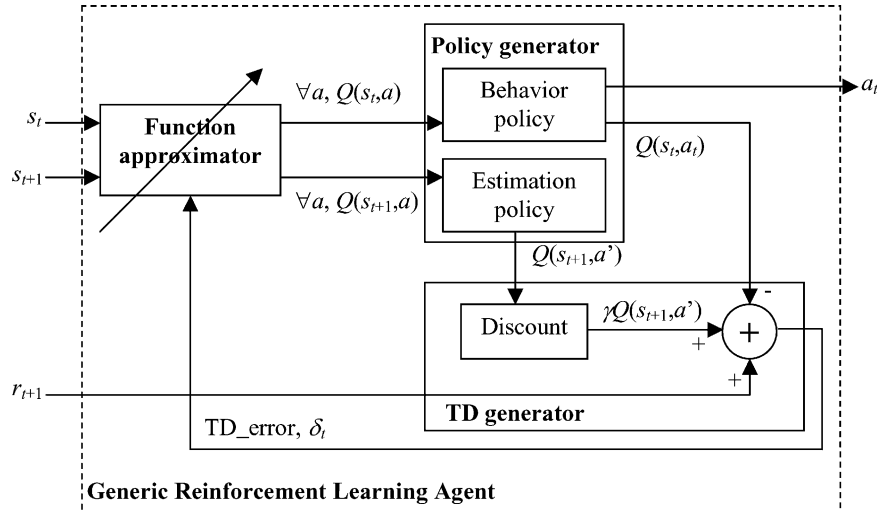


Fig. 1. A generic reinforcement learning framework.

are straightforward and will be discussed in Section 2.3. The differences in the behaviour and the strengths of the maximum reward reinforcement learning over existing reinforcement learning models will be demonstrated with simulations and examples in the latter part of this paper.

2.3. Mapping to a generic reinforcement learning framework

This subsection extends the maximum reward reinforcement learning update formulae (Eq. (2.15)) into the generic reinforcement learning framework proposed in (Quah & Quek, submitted for publication). The block diagram of the generic agent of the framework is shown in Fig. 1.

The generic agent adopts a modular design with three modules; namely: the function approximator, the policy generator, and the temporal difference (TD) generator. The function approximator module predicts the value functions based on the state information. The policy generator module generates an appropriate action and the corresponding action-value. The TD generator module subsequently formulates the temporal difference error based on the reinforcement signal, the current action-value, and the successive action-value. Consequently, the temporal difference error is feedback to the function approximator module to adjust the tunable parameters. Two motivations for having this modular design are the functional generality or abstraction, and the functional independence. Having generic functionality for each respective module enables the framework to be abstractly defined, and thus easily extendable to different reinforcement learning model such as TD learning, SARSA learning, or Q-learning. Moreover, the functional independence ensures that the modules are not tightly dependant on each other, and thus reformulation or implementation of a single module would not affect the general operations of the other modules. The discussion in this paper is limited to the following extent; the TD generator module is being generalized to incorporate the learning of maximum reward update formulae. Please refer

to (Quah and Quek, submitted) on the definitions of the function approximator and the policy generator modules.

The TD generator computes the temporal difference error as a function of the immediate reward, current action-value, and successive action-value estimation, as shown in Eq. (2.16).

$$\text{TD error : } \delta_t = f_{\text{TD}}(r_{t+1}, Q_t(s_{t+1}, a'), Q_t(s_t, a_t)) \quad (2.16)$$

where $f_{\text{TD}}(\cdot)$ is a generic function of the TD generator module.

The generic TD function is extended to the TD error of the cumulative reward formulation with Eq. (2.17).

$$\begin{aligned} \text{Cumulative reward TD error : } \delta_t &= f_{\text{TD}}(r_{t+1}, Q_t(s_{t+1}, a'), Q_t(s_t, a_t)) \\ &= r_{t+1} + \gamma Q_t(s_{t+1}, a') - Q_t(s_t, a_t) \end{aligned} \quad (2.17)$$

Similarly for the maximum reward formulation, the generic TD function adopts the formula as shown in Eq. (2.18).

$$\begin{aligned} \text{Non-cumulative(maximum)reward TD error : } \delta_t &= f_{\text{TD}}(r_{t+1}, Q_t(s_{t+1}, a'), Q_t(s_t, a_t)) \\ &= \max(r_{t+1}, \gamma Q_t(s_{t+1}, a')) - Q_t(s_t, a_t) \end{aligned} \quad (2.18)$$

The TD error formulae in Eqs. (2.17) or (2.18) can be easily mapped to TD learning, SARSA learning or Q-learning by adopting different action selection strategies in the policy generator module, as demonstrated in (Quah & Quek, submitted for publication). Furthermore, the maximum reward reinforcement learning can be directly incorporated into the FITSK-RL model (Quah & Quek, submitted) by adopting a zero-order FITSK model (Quah & Quek, submitted for publication) as the function approximator module. This enables the value function to be effectively approximated with a linear approximator (linear to the tunable parameter) in a systematic manner, and extendable to learning with the eligibility trace formulations (Sutton, 1984). Please refer to (Quah & Quek,

submitted for publication) on the technical details of formulating the FITSK-RL model.

3. Comparison with Q-learning and R-learning

This section compares the proposed non-cumulative, maximum reward reinforcement learning with existing learning models with inherent cumulative rewards reinforcement learning models, particularly the Q-learning (Watkins, 1989) and the R-learning (Schwartz, 1993). Q-learning is an off-policy, model-free reinforcement learning model, and is shown to be effective in control task learning with discounting of cumulative rewards. The learning equation for Q-learning is described by Eq. (2.14). The basic idea of formulating a Q-learning control is to adopt a greedy policy as the estimation policy, while simultaneously exploring the search space with a non-greedy behaviour policy. Similarly, R-learning is an off-policy, model-free model. However, the objective of R-learning is to maximize the performance based on the undiscounted cumulative rewards value, instead of the discounted cumulative rewards value described in Eq. (2.2). The undiscounted cumulative rewards value of a process starting at time t is shown in Eq. (3.1).

$$\rho_t = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^n r_{t+k+1} \quad (3.1)$$

where ρ_t is the average undiscounted cumulative rewards at time t . It was shown in (Schwartz, 1993) that if a policy gives rise to an *ergodic* Markov chain, the average reward values for all the states in that ergodic Markov chain are identical (Eq. (3.2)).

$$\rho = \rho_t, \quad \forall s_t \in S_E \quad (3.2)$$

where S_E is the set of Ergodic states.

The update formulae of R-learning are represented in Eq. (3.3). Please refer to (Schwartz, 1993) on the rational of these update formulae.

$$\begin{aligned} RV_{t+1}(s_t, a_t) &= RV_t(s_t, a_t) + \alpha_1(r_{t+1} - \rho_t \\ &\quad + \max_a RV_t(s_{t+1}, a) - RV_t(s_t, a_t)) \\ \rho_{t+1} &= \rho_t + \alpha_2 G(a_t)(r_{t+1} + \max_a RV_t(s_{t+1}, a) \\ &\quad - RV_t(s_t, a_t) - \rho_t) \end{aligned} \quad (3.3)$$

where,

$RV(\cdot)$ is a vector of realnumbers, corresponding to each state-action pair (s, a)
 α_1, α_2 are learning rate parameters, and
 $G(a_t)$ is a logic function with output 1 if a_t is greedily selected, and 0 otherwise.

Fig. 2 shows an example from (Mahadevan, 1996), which is an adaptation of the two cycles Markov decision process (MDP) task (Schwartz, 1993) with action choice at state 1 (Home) of the MDP. A robot following that MDP receives

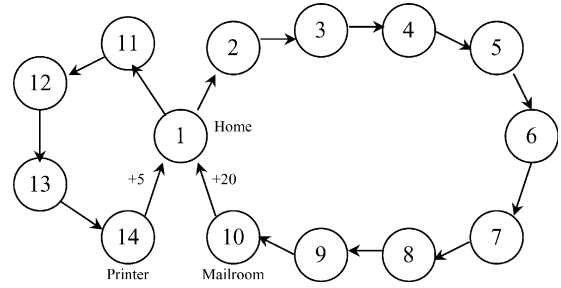


Fig. 2. An MDP with two cycles.

reward of +5 at the Printer room (state 14) for a trip from Home to Printer room, and receives reward of +20 for servicing the Mailroom (state 10) that is further away. Learning simulations are performed for R-learning (Eq. (3.3)), Q-learning (Eq. (2.14)) and Q_M -learning (Eq. (2.15)), respectively. All the learning rates have numerical value 0.2, and all the behavior policies are ϵ -greedy (Sutton, 1998) with $\epsilon=0.05$.

Fig. 3 shows the performance comparison between R-learning, Q-learning and Q_M -learning. The results are averaged over 5 independent runs, each run consisted of 20,000 number of steps. The performance is measured with the average of rewards being accumulated over the past 1000 number of steps. The optimal path follows the cycle from 1-2-3-4-5-6-7-8-9-10, and repeats itself infinitely. Therefore, the maximum of the average reward value is 2, and is derived by averaging the largest reward, 20, over 10 actions for each cycle. The learning agent may often stay at the suboptimal policy, which follows the cycle from 1-11-12-13-14, prior to arriving at the optimal policy. This suboptimal policy has an average reward of 1. A discount rate, $\gamma=0.9$, is adopted for both the Q-learning and the Q_M -learning. The discount rate in this MDP has to be larger than 0.7579 such that the policy of servicing the mailroom is being chosen. Any discount rate less than 0.7579 will cause a larger value at state 11 than the value at state 2. Thus, the policy servicing the Printer room will look more attractive than the policy servicing the Mailroom. R-learning arrived at the optimal solution in about 3700 number of steps, whereas Q_M -learning and Q-learning took 10,700 and 13,500 steps, respectively, to arrive at the optimal

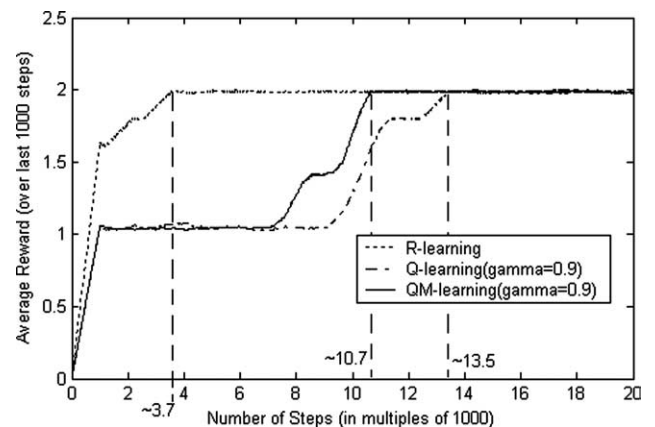


Fig. 3. Comparison of R-learning, Q-learning and Q_M -learning.

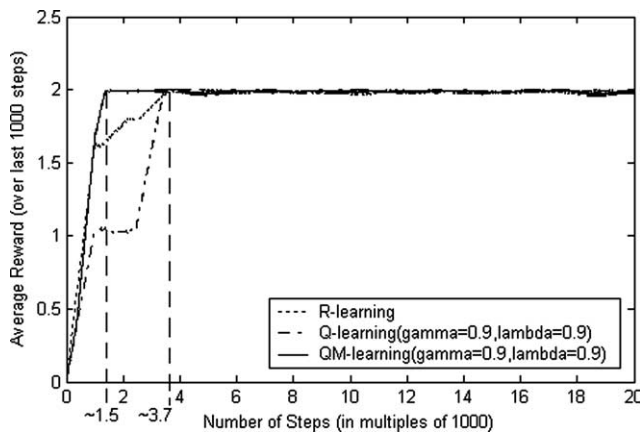


Fig. 4. Comparison of R-learning, Q-learning and Q_M-learning with eligibility traces.

solution. Comparing with R-learning in the undiscounted formulation, the learning with discounted formulation is prone to convergence to the suboptimal solutions. However, the maximum reward Q_M-learning performs slightly better than the cumulative reward Q-learning. The reason pertaining to this phenomenon will be discussed in the next simulation.

The reason that R-learning outperforming Q_M-learning in the previous simulation is investigated in Fig. 4. Eligibility traces formulations (Sutton, 1984; Sutton & Barto 1998) are introduced to the Q-learning and Q_M-learning with a large trace-decay parameter, $\lambda=0.9$. This ensures that the future rewards are apparent to the learning agent as soon as they are being discovered by a non-greedy exploratory policy. Fig. 4 shows that the Q_M-learning with eligibility trace has superior performance to the others. The Q_M-learning is more far-sighted than the others, by adopting the eligibility trace. Q_M-learning converges in about 1500 number of steps, while Q-learning and R-learning converges in about 3700 number of steps. The Q_M-learning converges to the optimal policy in the shortest possible timeframe without being dragged down by the suboptimal policy. This is consistent with the behaviour of maximum reward agent introduced in Section 2.2. The agent will attempt to discover the shortest path that lead to the highest reward, without concerning too much on the intermediate lower rewards. However, for cumulative rewards models such as Q-learning or R-learning, it takes time for them to fade away the influence of the lower reward and the effect of the suboptimal policy. Therefore, the convergence of these cumulative rewards models is not as immediate as the non-cumulative maximum reward model.

4. Optimal stopping problem in financial derivative pricing

In the original paper on average reward reinforcement learning (Schwartz, 1993), the author claimed that one of the reasons for discounting rewards is to compensate for the interest rate of future rewards, especially in the field of economic. This has been demonstrated by (Tsitsiklis & Van Roy, 1999; Van Roy, 1998) in the studies of an optimal stopping problem in financial derivative pricing. However, the

problem cannot be readily resolved by the discounted accumulative reward reinforcement-learning model. This is because the objective of the problem is to determine the optimal stopping state with the highest discounted non-cumulative rewards. This section attempts to demonstrate the application of the maximum reward reinforcement learning to model this financial derivative pricing problem.

4.1. Problem formulation

The optimal stopping problem was originally formulated by (Van Roy, 1998; Tsitsiklis, 1999) for optimal pricing prediction of an exotic derivative product (Parsley, 1997). The problem domain is a contract based on the daily prices of a single stock. When the contract is exercised, a payoff is received in an amount equal to the current price of the stock divided by the price a 100 days before. Please refer to (Tsitsiklis, 1999) on the background studies and details of this problem. The summary of this problem is given as follows.

The process of stock prices is generated based on a *geometric Brownian motion* model (Hull, 1997). The change in the stock price is modelled as a normal distribution with *drift rate* as the mean, and with *volatility* as the standard deviation, as described in Eq. (4.1).

$$\frac{\Delta p_t}{p_t} \sim \phi(\mu \Delta t, \sigma \sqrt{\Delta t}) \quad (4.1)$$

where

p_t is the stock price at time t ,

μ is the expected drift rate (percentage return, or interest rate of the stock price),

σ is the volatility (standard deviation of the stock price),

Δt is the interval of time, and

$\phi(m, s)$ is a normal distribution with mean m and standard deviation s

If the prices are computed on a daily basis, then the time interval is 1, and Eq. (4.1) is simplified into the update equation described in Eq. (4.2).

$$p_{t+1} = p_t(1 + \mu + \sigma \phi(0, 1)) \quad (4.2)$$

This problem considers the daily volatility of 0.02 and annual interest rate of 10%; the drift rate is 0.0004 by dividing the interest rate by the number of market opening days (~ 250 days). Therefore, the daily prices are updated based on Eq. (4.3).

$$p_{t+1} = p_t(1.0004 + 0.02\phi(0, 1)) \quad (4.3)$$

where $t \geq -100$.

The information available to the learning agent at a particular discrete time t is a vector, described as follows,

$$x_t = \left(\frac{p_{t-99}}{p_{t-100}}, \frac{p_{t-98}}{p_{t-100}}, \dots, \frac{p_t}{p_{t-100}} \right), \quad t \geq 0.$$

The payoff is rewarded if the contract is terminated at time t .

Therefore, the reward received a time step later is the last component of the information vector, $r_{t+1}=x_t(100)$. Then the value of the contract at time t is given by Eq. (4.4).

$$V_t = \sup_{\forall k \geq 0} E(\gamma^k r_{t+k+1}), \quad t \geq 0 \quad (4.4)$$

and the discount rate is given by, $\gamma = e^{-\mu}$. If t^* is an optimal stopping time, then the optimal value of the contract is described by Eq. (4.5).

$$V_{t^*} = E(r_{t^*+1}) = \sup_{\forall k \geq 0} E(\gamma^k r_{t^*+k+1}) \quad (4.5)$$

4.2. Learning model formulation and results

The state of the problem is constructed based on the following rationale. If there was a large slump and recovery in the stock prices within the past 100 days, then probably the contract should be held for some time to reap potential future benefits (Tsitsiklis, 1999; Van Roy, 1998). Therefore, the most important information is dependant on the current stock price, and the minimum stock price over the past 100 days. With this rationale, the state information is constructed as shown in Eq. (4.6). The first state dimension represents the proportional increment of the current stock price. The second state dimension implies the difference between the proportional increment of the current stock price over the minimum stock price of the past 100 days.

$$s_t = \left(\frac{p_t}{p_{t-100}}, \frac{p_t}{\min_{(t-100) \leq \tau \leq t} p_\tau} - \frac{p_t}{p_{t-100}} \right), \quad s_t \in \mathbb{R}^2 \quad (4.6)$$

In simulation trials, since the number of training iterations is always finite, there exist a maximum value in the set of the expected discounted payoff, and this maximum value is also the supremum of the set (Eq. (4.7)).

$$V_t = \max_{\forall k \geq 0} E(\gamma^k r_{t+k+1}) = \sup_{\forall k \geq 0} E(\gamma^k r_{t+k+1}), \quad (4.7)$$

$t \geq 0, \quad k$ is finite

Eq. (4.7) is equivalent to Eq. (2.6), which is the value function of the maximum reward reinforcement learning. Thus, this optimal stopping problem can be approximated with the maximum reward reinforcement learning, TD_M-learning, where the update error formula is shown in Eq. (2.15).

In this simulation, the optimal value function is approximated with the FITSK-RL model. The FITSK-RL model consists of a zero-order FITSK as the function approximator. The parameters for the learning are shown in Table 1.

The first and second dimensions are denoted as $s(1)$ and $s(2)$, which are computed using Eq. (4.6). The input ranges for the two state dimensions are $s(1) \in (0.5, 2.0)$ and $s(2) \in (0.0, 0.6)$, respectively. The number of pre-assigned membership

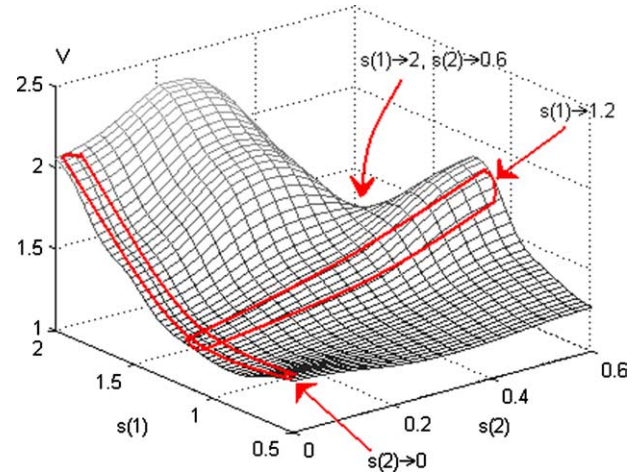


Fig. 5. Approximated optimal value function with maximum reward FITSK-RL.

functions for the two dimensions are 5 and 3. The total number of parameters to tune are $5 \times 3 = 15$. All the initial tunable parameters are set to zeros. The learning is iterated for 500,000,000 times. The approximated value function from the output of the model is plotted in Fig. 5.

Fig. 5 shows the surface plot of the approximated optimal value function with the proposed maximum reward FITSK-RL model after the learning iterations. The two state dimensions are represented with the X–Y axes, the approximated optimal value function is represented with the Z-axis. When the current stock price is near to the bottom of the past 100 prices ($s(2) \rightarrow 0$), then there is not much chance of getting better further benefits, and therefore expecting lower payoff. When the current stock price yields a growth of about 1.2 in $s(1)$, the expectation of payoff increases proportionally with $s(2)$. Increased in $s(2)$ means that there are larger slump and recovery over the past 100 days, and probably one should wait for sometime to reap potential higher payoff. The region at the upper right corner ($s(1) \rightarrow 2.0, s(2) \rightarrow 0.6$) are rarely encountered (see Fig. 5), and thus their learnt optimal value are lower than expected due to the lack of training examples. However, when such situation occurs, the process is being terminated immediately to reap the high benefits from these states. This is because the chance of getting any potential higher payoff than the current is quite slim.

Denoting the approximated optimal value function from the FITSK-RL as $\tilde{V}(s_t)$, then the stopping time, $\tilde{\tau}$, is computed by Eq. (4.8). This is similar to the terminating criterion defined by (Tsitsiklis, 1999), which was proven to be the optimal stopping time.

$$\tilde{\tau} = \min\{t | r_{t+1} \geq \tilde{V}(s_t)\} \quad (4.8)$$

The expected reward corresponding to the approximated optimal value function is estimated by averaging the rewards gathered over 50,000 independent trajectories, and terminated according to Eq. (4.8). The results are tabulated in Table 2. The expected reward computed from such stopping criterion is 1.289127, and the estimated standard deviation for this sample

Table 1

Predefined FITSK-RL parameters

Learning constant, $\alpha=0.001$	Discount rate parameter, $\gamma=e^{-\mu}=e^{-0.0004}$
Dimensions=2	Trace-decay parameter, $\lambda=0.5$

Table 2
Results of the optimal stopping problem

Strategy	Expected reward	Standard deviation	State dimension
Thresholding	1.238	–	–
Tsitsiklis model	1.282	0.0022	100
FITSK-RL	1.289	0.0029	2

mean is 0.002856. The expected reward is 1.238 from a thresholding strategy; and 1.282 (with 0.0022 standard deviation for sample mean) for the approximated optimal value function from (Tsitsiklis, 1999) formulation. Therefore, the expected payoff from the maximum reward FITSK-RL model is slightly better than the formulation by Tsitsiklis, which is leading by 0.007127. The expected reward is numerically 0.051127 better than the thresholding strategy. This means that the optimal pricing of the strategy identified by the maximum reward FITSK-RL model is 5% better than the thresholding strategy, and is about 1% better than the Tsitsiklis formulation.

4.3. Discussion

In the original paper on average reward reinforcement learning for the undiscounted reward problem, the author claimed that using discounting of rewards to compensate for the interest rate is unjustifiable (Schwartz, 1993). However, the optimal stopping problem in financial derivative pricing studied in this paper shows the contrary. The rewards are discounted by $e^{-\mu}$ to compensate for the possible future increment of the contract values, such that future values are comparable side-by-side with the current values. On the other hand, discounted cumulative reward reinforcement learning is not applicable to this problem. This is because the objective of the problem is to search for a highest expected reward for each state. Having this, a non-cumulative, discounted, maximum reward reinforcement learning formulation is shown to be directly applicable in solving this problem.

The FITSK-RL model is demonstrated to incorporate the maximum reward reinforcement learning formulation with ease in Section 2.3. The problem in this section is simulated with the maximum reward FITSK-RL model. The formulation presented in this section is slightly different to the one proposed in (Tsitsiklis, 1999). They formulated the problem with 100 state dimensions to demonstrate the good scaling of their local basis function approximator. However, the formulation in this paper reduces the state dimension to 2, such that the learning result can be easily visualized and inspected (as shown in Fig. 5). This improves the interpretability of the learning result. Furthermore, the operations of FITSK-RL model are systematically defined, as opposed to the manually generated local basis function approximation method in (Tsitsiklis, 1999). The expected payoff of the maximum reward FITSK-RL model with 2 state dimensions is slightly better than the existing methods with 100 state dimensions. This shows that the state definition in Eq. (4.6) with two dimensions can adequately provide enough information to the learning system. This is a

significant improvement in modelling complexity over the existing methods.

5. Conclusion

In this paper, a new maximum reward criterion is proposed as the objective of reinforcement learning. The maximum reward criterion is subsequently incorporated into the reinforcement learning update formulae from the Markov decision process perspective. Subsequently, the maximum reward reinforcement learning is mapped to the generic reinforcement-learning framework to solve the modelling issue of this new learning paradigm.

The distinct characteristic of the maximum reward reinforcement learning is a non-cumulative, maximum reward-oriented behaviour of the learning agent, as opposed to the usual cumulative rewards-oriented behaviour. It attempts to search for the policy that leads to the largest reward, irregardless of intermediate lower rewards. This positions it to have superior immunity to the existence of sub-optimal policies. This is demonstrated with the example in Section 3, where the maximum reward agent established a convergence to the optimal policy in the shortest timeframe. However, other agents with underlying cumulative rewards-oriented behaviour are being dragged down by the sub-optimal policy and lower reward. As a result, they suffer from slower convergence to the optimal policy.

Section 4 demonstrates a successful application of maximum reward reinforcement learning to asymptotically approximate the supremum of payoff values in an optimal stopping problem. The maximum reward FITSK-RL model proposed in Section 2.3 has been adopted to resolve this problem. This problem cannot be effectively solved using the usual cumulative rewards reinforcement-learning paradigm because the underlying payoff values are non-cumulative. Despite reducing the problem to a two-state dimension problem, the approximation result using the FITSK-RL model is slightly superior to the original formulation with 100 state dimensions in (Tsitsiklis & Van Roy, 1999). Furthermore, this improves the interpretability of the learning results and reduces the modelling complexity of the problem. The approximation result is visualized such that the rationale of the result can be examined with ease. This would not be possible with the original formulation of the problem as the state dimensions are too large and it is hard to gain insight into the learning result of the manually defined local basis functions. However, this is possible with the FITSK-RL model as the operations are systematically defined, and the state dimensions are small.

In summary, the maximum reward reinforcement learning approach is proposed as a possible new learning paradigm to exhibit superior performance for problems with underlying non-cumulative nature of reward values. Examples and simulations are provided to illustrate this. The modeling issue of the maximum reward reinforcement learning is solved by mapping it into the generic reinforcement learning framework. However, the theoretical research of maximum reward reinforcement learning is still at its infancy stage, particularly from the optimal control perspective. The convergence of maximum reward reinforcement learning

resulted from a linear function approximator is yet to be established. Some possible resemblance of the convergence result may possibly be extended from the Tsitsiklis' papers (Tsitsiklis & Van Roy, 1997, 1999), which justifies a need of possible further research. The relationship to the sensitive discount optimality criterion (Mahadevan, 1996; Puterman, 1994) has not been studied, and is worthy of a further research effort.

References

- Bertsekas, D. P., & Tsitsiklis, J. N. (1996). *Neuro-dynamic programming*. Belmont, MA: Athena Scientific.
- Hull, J. C. (1997). *Options, futures, and other derivatives* (3rd ed.). Englewood Cliffs, NJ: Prentice Hall.
- Kaelbling, L. P., Littman, L. M., & Moore, A. W. (1996). Reinforcement learning: A survey. *Journal of Artificial Intelligence Research*, 4, 237–285.
- Mahadevan, S. (1996). Average reward reinforcement learning: Foundations, algorithms, and empirical results. *Machine Learning*, 22, 159–196.
- Parsley, M. (1997). Exotics enter the mainstream. *Euromoney*, 127–130.
- Puterman, M. L. (1994). *Markov decision processes: Discrete stochastic dynamic programming*. New York, NY: Wiley.
- Quah, K. H., & Quek, C. (2006). FITSK: Online local learning with generic fuzzy input Takagi-Sugeno-Kang fuzzy framework for nonlinear system estimation. *IEEE Transactions Systems, Man, and Cybernetics, Part B: Cybernetics*, 36(1), 166–178.
- Quah, K. H., & Quek, C. (submitted for publication). FITSK-RL: Function approximation with the fuzzy input Takagi-Sugeno-Kang model for a generic reinforcement learning framework.
- Rummery, G. A., & Niranjan, M. (1994). On-line Q-learning using connectionist systems, *Technical report CUED/F-INFENG/TR166*, Cambridge University Engineering Dept. http://svr-www.eng.cam.ac.uk/reports/svr-ftp/rummery_tr166.ps.Z.
- Schwartz, A. (1993). A reinforcement learning method for maximizing undiscounted rewards, *ICML 1993*, pp. 298–305.
- Sutton, R. S., (1984). *Temporal credit assignment in reinforcement learning*, PhD dissertation, University of Massachusetts, Amherst, MA.
- Sutton, R. S. (1988). Learning to predict by the method of temporal differences. *Machine Learning*, 3(1), 9–44.
- Sutton, R. S., & Barto, A. G. (1998). *Reinforcement learning: An introduction*. Cambridge, MA: The MIT Press.
- Tadepalli, P., & Ok, D.-K. (1998). Model-based average reward reinforcement learning. *Artificial Intelligence*, 100, 177–224.
- Tsitsiklis, J. N., & Van Roy, B. (1997). An analysis of temporal-difference learning with function approximation. *IEEE Transactions on Automatic Control*, 42(5), 674–690.
- Tsitsiklis, J. N., & Van Roy, B. (1999). Optimal stopping of Markov processes: Hilbert space theory, approximation algorithms, and an application to pricing high-dimensional financial derivatives. *IEEE Transactions on Automatic Control*, 44(10), 1840–1851.
- Van Roy, B. (1998). *Learning and value function approximation in complex decision processes*, PhD dissertation, Massachusetts Institute of Technology.
- Watkins, C. J. C. H. (1989). *Learning with delayed rewards*, PhD dissertation, Psychology Department, Cambridge University, England.